

Habit Creator

An end-to-end product research and UX case study on helping users build and sustain habits

Case-study thesis

The core problem was not getting people to start a habit. The research consistently pointed to a harder challenge: sustaining behavior after initial motivation fades. The product direction therefore focused on visible progress, low-friction interaction, adaptable motivation, and supportive social mechanics without harmful comparison.

Project context	Scope
Course	Human–Computer Interaction (HCI), University of Isfahan
Team	4-person academic product/design team
My positioning	Product research, competitive analysis, UX/product strategy, early product design
Research	10+ semi-structured interviews + 24-response survey
Product work	Personas, problem statements, competitor audit, ideation, prioritization, IA, user flows, wireframes, low-fidelity prototype
Evaluation	Scenario-based prototype evaluation framework; 6 simulated participant profiles; task metrics, SUS and NPS

Prepared as a polished English portfolio document from the original Phase 1 and Phase 2 HCI project documentation.

1. Executive Overview

Habit Creator is a mobile habit-building concept designed for users who repeatedly start positive routines or attempt to stop unwanted behaviors, but struggle to maintain change over time. The project followed a research-led HCI process: understand real behavior, synthesize recurring patterns, compare existing solutions, define product opportunities, generate and prioritize concepts, design user flows and a low-fidelity prototype, then evaluate the main interaction paths.

The problem we set out to understand

- Why do users abandon habit-change efforts after an initially motivated start?
- Which motivational mechanisms actually help—personal progress, rewards, social support, competition, or environmental cues?
- Where do current habit-tracking products create friction or fail to support long-term motivation?
- How can a product create social support without turning the experience into stressful comparison?

Research / product dimension	What was done
Foundational research	Semi-structured interviews grounded in The Mom Test; focus on past behavior rather than hypothetical intent.
Qualitative synthesis	Open coding / thematic analysis to identify repeated behavioral patterns and pain points.
Quantitative validation	Online survey with 24 responses, using closed/Likert and semi-open questions.
Behavioral modeling	Empathy maps, 3 behavioral personas, user stories and user journeys.
Competitive intelligence	5 direct habit competitors + 4 indirect alternatives assessed against a common framework.
Product definition	Problem statements, user goals, constraints, goal statement and success metrics.
Ideation & prioritization	HMW, Reverse Thinking, Radical Change, Turn Negative into Positive, Crazy Eights, 30 ideas and value-vs-complexity prioritization.
Information design	Open card sorting, information architecture, 3 key user flows and a storyboard.
Prototype & evaluation	Figma wireframes + clickable low-fidelity prototype; scenario-based evaluation using task success, time-on-task, error points, drop-off risk, SUS and NPS.

2. Phase 1 — Foundational User Research

The first phase was intentionally problem-focused. Rather than starting with a feature concept, the team investigated how people actually attempt to build, maintain, or stop habits, what causes them to relapse, and what existing tools they use in the process.

2.1 Interview approach: The Mom Test

The interview guide was designed to reduce confirmation bias. Questions referred to concrete past behavior, recent attempts, actual tools, specific moments of failure, and real emotional or social contexts. Hypothetical questions such as “Would you use an app that...?” were intentionally avoided.

- Semi-structured format allowed participants to explain context freely while keeping the discussion anchored to comparable themes.
- Questions targeted triggers, duration of adherence, drop-off moments, obstacles, emotions, tool usage, social influence, competition and intrinsic/extrinsic motivation.
- More than 10 users were interviewed; the project documentation describes this sample as sufficient to reach recurring qualitative patterns / data saturation.

2.2 Core interview framework

Research area	Purpose
Daily context	Walk through a typical day; identify energy peaks, fatigue, attention demands and habit windows.
Past behavior change	Explore real attempts to build or stop habits and the trigger behind the attempt.
Recent example	Anchor discussion in the latest concrete behavior-change attempt.
Start trigger	Understand whether the change was planned, spontaneous, self-driven or socially influenced.
Adherence duration	Measure how long the behavior lasted and when continuation became difficult.
Drop-off point	Identify the exact moment and context where abandonment was most likely.
Pain points	Separate internal barriers (motivation, emotion) from external barriers (time, environment, access).
Emotional state	Understand guilt, pressure, confidence, frustration and their effect on decisions.
Current tools	Identify direct and indirect substitutes, why they were chosen, and why usage decreased.
Social environment	Assess support, judgment, peer influence and preference for solo vs. shared progress.
Competition / group activity	Test whether competition motivates, stresses, or has context-dependent effects.
Motivational drivers	Distinguish intrinsic progress/reward from external approval.
Sharing behavior	Understand when users share achievements and whether controlled sharing would reduce friction.

2.3 Representative behavioral patterns from interviews

The raw notes captured varied contexts—screen time, exercise, study routines, sleep, smoking, meditation and other repetitive behaviors. For portfolio use, the personally identifying names are omitted and the findings are presented as behavioral patterns.

Pattern	Evidence synthesized from interview notes
Immediate-reward dependence	Some participants understood that excessive phone/social-media use harmed their well-being, yet still returned to it because of immediate reward and lack of an attractive replacement activity.
Environment-driven behavior	Smoking, hookah use, phone use and exercise adherence were often strongly shaped by place, availability and social groups. Changing the environment sometimes changed behavior more effectively than willpower.
Motivation decay	Participants frequently started with strong intention, but the initial excitement faded after days or weeks.
Comparison pressure	For several users, seeing others perform better did not motivate them; it created pressure, inadequacy or overexertion.
Visible progress	Streaks, body changes, completed sessions and visible history could reinforce continuation when progress felt tangible.
Need for variety	Some users became bored by fixed routines and responded better to variety, changing environments and progressive difficulty.
Low-friction tools	Simple phone notes and informal tracking were common, suggesting that complexity itself can become a barrier.
Support without pressure	Group rules or shared activity could help some users, but public ranking or forced comparison was not universally beneficial.

3. Qualitative Synthesis — Thematic Analysis

After the interviews, qualitative data were analyzed using open coding. Key statements were grouped by semantic similarity to surface recurring themes and translate research into concrete design implications.

Theme	Research insight	Product implication
Failure happens in continuation, not initiation	Most users can start; many stop after initial excitement fades.	Design for sustained motivation and relapse recovery—not only onboarding.
Behavior is highly contextual	Social environment, place and routine can influence behavior more strongly than individual intention.	Use context-sensitive prompts and flexible routines; do not assume willpower is sufficient.
Social comparison can backfire	Competition caused stress or feelings of inadequacy for several users.	Make competitive mechanics optional; prefer supportive or low-pressure social validation.
Visible progress matters	Streaks and visible personal progress can support motivation.	Keep progress feedback prominent, easy to understand and frequently available.
Static routines become boring	Some users lose motivation when a routine remains repetitive.	Use gradual levels, variety or adaptable challenges to preserve novelty.

Most important research insight

The dominant failure mode was not “I do not know how to start.” It was “I start, but I cannot keep going once motivation drops.” That changed the product focus from initiation to sustained engagement, progress visibility and recovery after lapses.

4. Quantitative Validation — Online Survey

The survey was used to test whether the qualitative patterns appeared across a broader sample. It targeted mainly students and young users familiar with lifestyle or habit-management tools. The questionnaire combined closed/Likert-style questions with semi-open responses.

4.1 Survey scope

- Attempts to create or stop habits
- Frequency of interruption or failure
- Intrinsic motivation, competition and social validation
- Effect of visible progress and streaks
- Attitudes toward sharing progress and achievements

Finding	Reported result / interpretation
Existing demand	More than two-thirds of respondents had previously attempted to create or stop a habit, indicating an experienced—not hypothetical—problem.
Main difficulty	About 45% identified maintaining consistency as the hardest part, materially higher than the share identifying initiation as the main challenge.
Repeated failure pattern	More than half reported that obstacles to behavior change recur, suggesting a systemic rather than one-off problem.
Tracking burden	Most respondents rated progress tracking as moderately to highly difficult, indicating cognitive overhead in habit management.
Motivation	Intrinsic satisfaction and visible personal progress were reported more often as motivators than competition or social comparison.
Current substitutes	Users often relied on simple tools such as phone notes rather than specialist habit apps.
Time-of-day risk	Late-day periods were reported as the most likely time for abandoning plans, consistent with fatigue / lower energy.

Survey sample: 24 responses. The original project documentation uses these results to validate the qualitative findings and guide product direction.

5. From Research to Personas and Problem Statements

The team created three behavioral personas from recurring research patterns rather than demographic assumptions. Each persona represents a different motivational mechanism and failure mode, giving the product team a decision framework for design trade-offs.

Behavioral persona	Core problem	Primary design need
Sadaf — progress & validation	Disciplined, goal-oriented, but vulnerable to perfectionism and motivation loss when effort is not visible.	Needs continuous visible progress, fast feedback, controlled social validation and small-step recovery after lapses.
Roza — emotion & novelty	Motivation changes with mood; repetitive long-term routines become boring.	Needs variety, gradual progression, visible progress and low-pressure ways to keep moving on low-energy days.
Kourosh — environment & support	Strongly influenced by peers and environment; social support helps him begin, but comparison can create inadequacy.	Needs a supportive start, lightweight external triggers and a path toward independent self-efficacy without harmful competition.

5.1 Example problem statements

- Sadaf needs a way to see progress continuously and receive controlled feedback because invisible effort and lack of recognition reduce motivation and increase relapse risk.
- Roza needs a way to keep long-term routines from becoming emotionally flat and repetitive because her commitment falls when novelty and immediate motivation fade.
- Kourosh needs a way to begin with social support and gradually become more independent because he benefits from companionship but can be harmed by direct comparison and competitive pressure.

5.2 Design constraints identified

Constraint area	Design requirement
Technical	Mobile-first (iOS / Android), lightweight experience, usable on mid-range devices; low-data / partial offline support where possible.
Time / MVP	Prioritize essential flows and defer complex features because the prototype had a limited academic development window.
Budget / resources	Avoid dependence on costly infrastructure or advanced services for the initial concept.
Privacy / ethics	Give users control over personal behavior data and sharing; avoid designs that create guilt, shame or psychological pressure.
Visual / brand	Minimal, understandable, low-cognitive-load interface; avoid dark patterns or excessively stimulating engagement mechanics.
Accessibility	Readable typography, adequate contrast, clear structure and usability under low-attention / fatigue conditions.

6. Competitive Analysis

The competitor audit examined both direct habit-tracking products and indirect substitutes. A common evaluation framework was used so that strengths, weaknesses and market gaps could be compared consistently rather than as isolated app reviews.

6.1 Competitive set

- Direct: Habitica, Fabulous, Routino, Streaks, Way of Life
- Indirect: Todoist, Notion, Google Calendar, paper planners / manual checklists

6.2 Evaluation criteria

Dimension	Questions used in the audit
Target & value proposition	Who is the product for? What core promise does it make?
First impression / UX	Does first use feel motivating, simple and trustworthy or confusing and overloaded?
Interaction model	How intuitive are core flows? Which reminders, progress, gamification and social features exist?
Navigation & accessibility	Are menus and paths clear? Can a broad range of users access the product? Is Persian supported?
Visual design	Is UI consistent with brand and behavioral context? Are icons, charts and hierarchy useful?
Content / tone	Is the tone supportive, professional, playful or overly dense? Is information easy to locate?
Strengths, weaknesses, opportunities	Which mechanisms work well, which create friction, and where can a new product differentiate?

6.3 Direct competitor audit

Competitor	Strength / experience	Key weakness
Habitica	Highly engaging, game-like experience with strong gamification and colorful character mechanics.	High complexity and learning cost, especially for non-gamers; can feel visually busy.
Fabulous	Professional, motivational, story-led and step-based experience with calm minimalist design.	Heavy reliance on long-form content and limitations in the free version.
Streaks	Simple, polished habit continuity tracking with strong focus on streaks and low-friction interaction.	iOS-only positioning in the project analysis and limited social functionality.
Routino	Simple, Persian-language routine and habit management with reminders and progress statistics.	Limited gamification and a comparatively basic / dated visual experience.
Way of Life	Organized, data-oriented tracking with daily logging, reminders and weekly/monthly trend analysis.	Functional but emotionally dry; limited motivational or visually engaging mechanics.

Competitive gap

The audit suggested a space between simple local tools and richer international products: a Persian-first, low-friction habit experience that combines visible progress and selective gamification with supportive social mechanics—without importing unnecessary complexity or pressure.

6.4 Opportunities identified

- Combine the simplicity of local products with the motivational quality and visual polish of stronger international products.
- Support both habit formation and habit reduction / stopping behavior.
- Use concise, personalized motivational content instead of long, cognitively heavy content.
- Make continuous interaction simple enough to reduce abandonment.
- Use community / social support as an optional motivational layer rather than mandatory public competition.
- Blend gamification and visual appeal with a minimal core experience.

7. Ideation and Feature Prioritization

Research insights were converted into design opportunities using How Might We questions. To widen the solution space, the team also used Reverse Thinking, Radical Change and Turn Negative into Positive, followed by Crazy Eights. This produced 30 concepts across the three personas.

7.1 Examples of research-to-feature translation

Observed need	Design direction explored
Progress must remain visible	Persistent progress / streak surfaces, widgets, milestone feedback and visible achievements.
Long-term repetition reduces motivation	Progressive levels, varying challenges, adaptive experience and novelty.
Social validation can help but comparison can hurt	Small supportive groups, buddy models, controlled sharing, non-ranked social signals.
Users may quit after one bad day	Grace-day / relapse-recovery concepts to avoid an all-or-nothing failure experience.
Identity can reinforce behavior	Personalizable character / profile evolving with progress.
Low-energy moments need small interventions	Lightweight external triggers, reminders and low-cost motivational nudges.
Users can feel alone without wanting a community	Ambient signals such as “others are doing this now” without requiring direct participation or ranking.

7.2 Prioritization matrix

The 30 ideas were compared on two axes: User Value and Implementation Complexity. Similar concepts were merged, and higher-risk or low-value features were deferred. The output created an explicit MVP logic instead of allowing the feature set to expand arbitrarily.

Quadrant	Definition	Use
Easy Wins	High user value, lower implementation complexity	Best candidates for initial product / MVP.
Big Bets	High user value, higher implementation complexity	Strategically important, but more suitable for later or selective inclusion.
Fill-ins	Lower complexity, moderate value	Useful complementary features.
Avoid / Later	Low value, poor value-to-effort ratio, or elevated UX risk	Defer or exclude.

- Easy Wins plus selected Big Bets became the foundation for the information architecture, wireframes and prototype.

Figma prioritization matrix: [Open board](#)

8. Product Goal and Success Metrics

Goal statement

Help users build and maintain daily and long-term habits by making progress tangible and reinforcing motivation through supportive social validation. Success should be measured through consistency, habit completion, engagement and user satisfaction—not only app usage.

8.1 Proposed success metrics

Area	Metrics / signals
Usability	Time on task, error rate, task completion rate.
Engagement	Daily/weekly habit check-ins, meaningful interaction frequency, average usage where relevant.
Behavior continuity	7-day / 14-day / 30-day continuation, streak continuity, relapse recovery.
Retention	Share of users remaining active after week one; reduced drop-off after motivation dips.
Satisfaction	Perceived control, motivation, overall UX satisfaction, SUS / similar usability measures.
Psychological outcome	Reduced guilt / self-blame and increased sense of achievement from small progress.
Goal alignment	Does the experience visibly sustain motivation and allow sharing without social pressure?

9. Information Architecture and User Flows

9.1 Open card sorting

Open card sorting was selected to derive information structure from users' mental models rather than forcing the team's preferred taxonomy. Participants grouped functions such as character creation, habits, today's tasks, challenges, achievements, friends, sharing, rewards and notifications into categories that felt natural to them.

- Purpose: discover intuitive categories and labels.
- Outcome: inform navigation, reduce search friction and align menu structure with users' expectations.
- Example card set: create character, first habit, today's tasks, active challenges, create good habit, stop bad habit, customize habit, completed activity, points, friends, profile, achievements, notification choice, share achievement, level up, reward items, follow others.

9.2 Core user flows

1. Create / complete a habit and receive progress or reward feedback.
2. Share activities and achievements.
3. Create a new habit, including either building a positive habit or stopping an unwanted one.

Figma user flows / wireframes: [Open design](#)

10. Storyboard, Wireframes and Low-Fidelity Prototype

10.1 Big-picture storyboard

A storyboard based on the progress-and-validation persona mapped the experience from a realistic failure moment—fatigue leading to social-media scrolling—through product discovery, profile creation, habit setup, visible progress, community support and sharing. The purpose was to make the product value proposition concrete before investing in UI detail.

Storyboard stage	Role in the experience
Everyday context	User is tired and defaults to an easier immediate-reward behavior.
Failure / frustration	User recognizes another missed attempt and feels dissatisfied.
Product entry	The habit app offers a more structured and motivating path.
Personalization	Character/profile creates ownership and identity.
Progress & streak	Visible feedback makes effort tangible.
Support	Community / friends provide controlled encouragement.
Continuation	Progress, feedback and support reduce all-or-nothing abandonment.

10.2 Wireframes and prototype

- Wireframes in Figma covered the main screens, user flows and decision points before visual polish.
- A clickable low-fidelity prototype connected the key flows to enable early interaction evaluation.
- The prototype intentionally prioritized structure and behavior over final UI styling so that navigation and task logic could be evaluated cheaply.

11. Prototype Evaluation

The Phase 2 document defines a scenario-based usability evaluation around the Figma prototype. It uses six simulated participant profiles and estimated near-realistic task times, alongside simulated SUS and NPS responses. This makes the section best understood as a structured prototype-evaluation exercise rather than evidence from a production usability study.

11.1 Research questions

- Can users sign up and create / customize a profile or character without difficulty?
- Can users create a reading habit and record completion?
- Do users clearly understand progress, streaks and rewards?
- Can users find and use sharing / social interaction (friends or community)?
- Does the overall experience feel motivating, engaging and sustainable?

11.2 Tasks and results

Task	Avg. time	Range	Task success
Sign-up & profile	2:15	1:40–3:20	83%
Create reading habit	1:35	1:05–2:10	100%
Record habit completion	0:55	0:35–1:20	100%
View progress / streak	1:20	0:50–2:05	83%
Share achievement	1:50	1:10–2:40	66%
Enter community / interact	2:05	1:30–3:15	66%

Evaluation signal

The core habit flows were straightforward, while the social and sharing paths were slower and had lower success. The concept therefore needed clearer social information architecture before further visual refinement.

11.3 Key usability issues

Observed issue	Implication / redesign direction
Share action was hard to find	Move sharing closer to the progress / achievement context where users naturally expect it.
“Community” vs. “Friends” was unclear	Clarify the relationship model, labels and navigation hierarchy.
Progress was not prominent enough	Surface progress / streak more clearly on the home experience.
Character customization felt too long	Reduce choices or defer advanced customization; keep first-use setup fast.
Potential long-term drop-off	Two profiles indicated they might stop if personalization became lengthy or feedback / reminders were insufficient.

11.4 SUS and NPS (from the simulated evaluation)

Participant	SUS	NPS classification
Participant 1	82	9 — Promoter
Participant 2	74	7 — Passive
Participant 3	80	8 — Passive
Participant 4	78	9 — Promoter
Participant 5	85	10 — Promoter
Participant 6	68	6 — Detractor

- Calculated average SUS from the documented values: 77.8 / 100.
- Calculated NPS from the documented classifications: approximately +33 (50% promoters minus 16.7% detractors).
- Interpretation: the concept’s basic usability was promising, but social navigation and sharing required iteration before a stronger validation round.

12. How Research Changed the Product

The strongest part of this project is the traceability between evidence and product decisions. Research was not treated as a separate deliverable; it directly shaped the concept, feature logic, prioritization and evaluation criteria.

Evidence	Decision
Users fail after motivation fades	Prioritize sustained progress feedback, relapse recovery and long-term motivation.
Progress visibility is motivating	Make streaks, milestones and progress central—not buried analytics.
Competition can create stress	Avoid mandatory leaderboards; use supportive, optional social mechanics.
Environment strongly affects behavior	Consider contextual triggers and adaptable routines rather than relying on willpower.
Simple substitutes are common	Keep the core interaction low-friction and avoid feature overload.
Repetition reduces motivation for some personas	Add controlled novelty, levels or varying challenges.
Social paths underperformed in prototype evaluation	Clarify Friends vs. Community, reposition Share and simplify the social IA.

13. What This Project Demonstrates for Product / Research Roles

Capability	Evidence from the project
User research	Interview design, The Mom Test principles, semi-structured qualitative research, survey design.
Research synthesis	Open coding, thematic analysis, insight extraction, quantitative/qualitative triangulation.
Competitive intelligence	Structured competitor framework, direct/indirect competitor mapping, gap identification.
Product discovery	Problem statements, personas, HMW, value proposition and opportunity framing.
Feature strategy	30-idea solution space, user-value vs. complexity prioritization, MVP thinking.
UX / HCI	Empathy maps, user stories, journeys, card sorting, IA, flows, wireframes and prototyping.
Product metrics	Task success, time on task, error/drop-off thinking, retention, engagement, SUS and NPS.
Decision communication	Clear chain from user evidence → insight → design opportunity → prototype → evaluation.

14. Resume-Ready Version

SELECTED ACADEMIC PRODUCT RESEARCH PROJECT — HABIT CREATOR

University of Isfahan | Human–Computer Interaction | Product Research & UX Strategy

- Conducted end-to-end product discovery for a habit-building mobile concept, combining 10+ semi-structured user interviews and a 24-response survey to identify recurring behavioral barriers and validate product needs.
- Applied The Mom Test principles and thematic analysis / open coding to identify five core behavioral patterns, including motivation decay, environmental influence, negative social comparison, need for visible progress, and fatigue from repetitive routines.
- Built three behavior-based personas, empathy maps, problem statements, user stories and journeys to translate qualitative and quantitative insights into design requirements.
- Led / contributed to a structured competitor audit covering 5 direct and 4 indirect alternatives, evaluating value proposition, UX, features, visual design, content, accessibility, strengths and market gaps.
- Generated and evaluated 30 solution concepts using HMW, Crazy Eights and divergent-thinking methods, then prioritized features using a User Value × Implementation Complexity matrix to define MVP direction.
- Developed information architecture through open card sorting, mapped three core user flows, and created Figma wireframes and a clickable low-fidelity prototype.
- Designed a scenario-based prototype evaluation framework using task success, time on task, error points, drop-off risk, SUS and NPS; analysis highlighted social navigation and sharing as the highest-priority usability issues.

Keywords for ATS / LinkedIn

Product Research • Competitive Analysis • Market / Competitor Intelligence • User Research • Product Discovery • HCI • UX Research • Thematic Analysis • Survey Design • Personas • Problem Framing • Feature Prioritization • MVP Strategy • Information Architecture • User Flows • Wireframing • Prototyping • Usability Evaluation • Product Metrics • Figma

15. Interview Talking Points

Question you may be asked	Strong answer direction
“Tell me about a research project.”	Explain how you started with real behavior rather than a solution, used interviews + survey triangulation, and discovered that consistency—not initiation—was the core problem.
“How did competitive analysis influence the product?”	Describe the gap between simple local tools and richer but more complex international apps, then connect that gap to the chosen design direction.
“How did you prioritize features?”	Explain the 30-idea pool and the User Value × Implementation Complexity matrix, including Easy Wins vs. Big Bets.
“Give an example of research changing a design decision.”	Use the negative-comparison insight: social features became supportive / optional rather than leaderboard-first.
“How did you evaluate the prototype?”	Be precise: the course document used a simulated scenario-based evaluation framework with task metrics, SUS and NPS; use it to show method knowledge without presenting it as live production research.

16. Project Artifacts

Prioritization matrix: [Figma board](#)

Wireframes / user flows / low-fidelity prototype: [Figma design](#)

Source basis: Phase 1 HCI research documentation and Phase 2 Habit Creator product-design documentation supplied by Arian Jafari. This English version is edited for clarity, professional terminology, portfolio readability and defensible product/research positioning; it is not a literal line-by-line translation.